

Hints for Assignment VI

- 1-9 Trivial. Recall that if T is an operator from V to W where $\dim(V) = n$ and $\dim(W) = m$, then matrix of T will be of order $m \times n$.
- 10 (a) Write $A = GG^*$, where G is an invertible matrix.
(b) Trivial.
- 11 (a) Trivial.
(b) Use the definition of transpose of a matrix. $\langle Ax, y \rangle = \langle x, A^T y \rangle$. Suppose $Ax = \lambda x$. Then $\langle Ax, y \rangle = \lambda \langle x, y \rangle$. But by the definition of transpose, what is LHS?
(c) How was the characteristic polynomial defined?
- 12 Produce a non-zero vector x such that $Ax = \alpha x$.
- 13 Trivial.
Yes, the eigenvalues of $p(T)$ are of the form $p(\lambda)$. How?
- 14 Trivial.
- 15 Is it necessary for S and T to have the same eigenvector corresponding to λ and μ respectively?
- 16-19 Trivial.
- 20 (a) Show $\lambda = \bar{\lambda}$, for all eigenvalues λ .
(b) Use the definition of orthogonal vectors.
(c) Realise that $x \times y$ is perpendicular to both x and y and since x and y are eigenvectors corresponding to distinct eigenvalues, by (b), they are orthogonal to each other. We are in $\mathbb{R}^3$ and we have three vectors perpendicular to each other. From here, how will you show that $x \times y$ is an eigenvector?
- 21 Use the fact that $A^*A = AA^* = I$ and the definition of adjoint of a transformation.